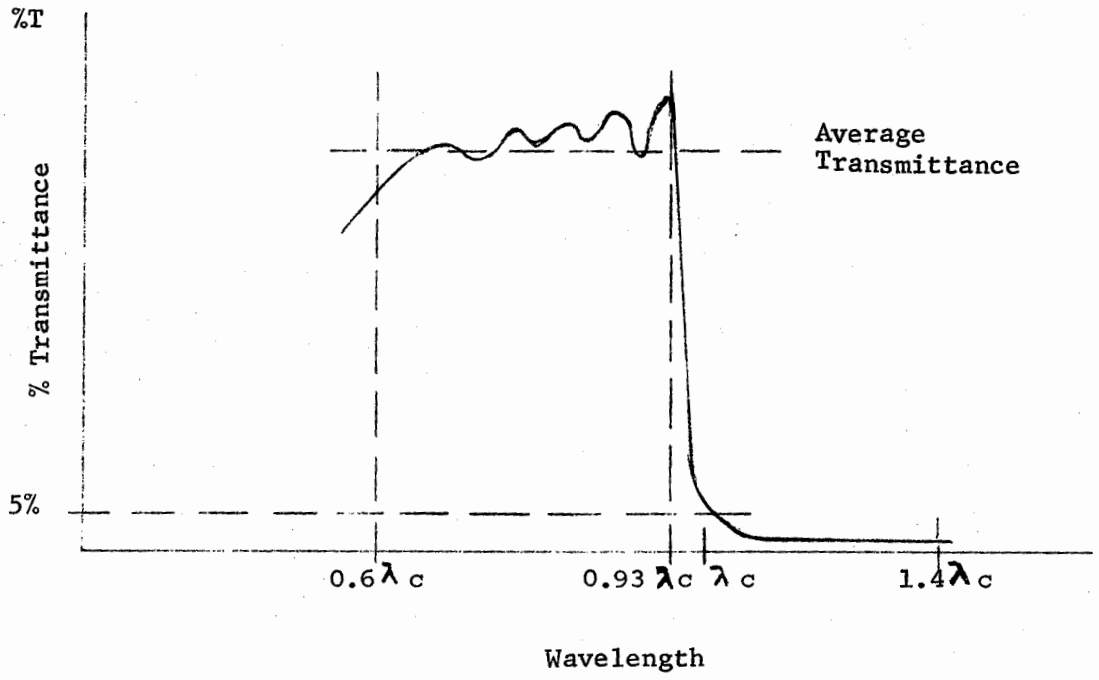
MILITARY SPECIFICATION SHEET

COATING, SHORT WAVELENGTH PASS
FILTER, ORDINARY REQUIREMENTS FOR

The complete requirements for procuring the short wavelength pass filters described herein shall consist of this document and the latest issue of MIL-F-48616.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The following figure illustrates the terms used for describing short wavelength pass filters.



NOTES FOR FIGURE:

1. The cutoff wavelength (λ_c) is the wavelength at which the absolute transmittance is 5%.

2. Average transmittance is determined between 0.6 λ_c and 0.93 λ_c.

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Short Wavelength Pass Filter Spectral Requirements

TABLE I

Cut-Off Wavelength (λ_c)	Minimum Average Transmittance	Dash No.
0.9 to 15.0 um	75%	1
15.0 to 25.0 um	70%	2
25.0 to 35.0 um	60%	3
35.0 to 50.0 um	50%	4

NOTES ON SPECTRAL REQUIREMENTS:

1. The 5% absolute transmittance point shall be located to within ±2% of the specified wavelength.

2. Transmittance shall be equal to or less than 0.1% absolute from 1.07 λ_c to 1.4 λ_c .

3. The values in this table apply at an angle of incidence ranging from 0^o to 5^o and at ambinet temperature. Measurements will be made at an angle of incidence within this range (See 6.4b and 6.4c).

4. At wavelengths longer than 1.4 λ_c short wavelength pass filters may have high transmittance.

5. Spectral resolution to verify the above requirements in the transmitting and cut off regions shall be 0.5% or less of the cut off wavelength. Resolution shall be 10% or less of the cut off wavelength in the attenuation region. Scan speed and time constant shall be set such that the spectral features of interest can be resolved within the accuracy of 4.4.2.3.2.2 of MIL-F-48616.

Ordering Data:

At a minimum the cut off wavelength (See 6.2m) (λ_c) and the dash number (See 6.2b) shall be specified by the procuring activity.

Preparing activity:

Army - MU

Project No. 6650-0091-7

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